PHAN VAN BANG

Trading Systems Developer · MQL5, Algo Trading, Full-Stack

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SUMMARY

Software engineer who designs and ships algorithmic trading systems end to end. Built **B. Nexus Quantum**, a commercial MetaTrader 5 Expert Advisor of ~27,000 lines of MQL5 in 11 modules with 21 entry modes, an 18-layer scoring engine and 14+ layers of risk control, plus its licensing and payment website (Next.js, Prisma, PayOS). Comfortable moving between MQL5, Python (Binance API, Telegram bots, TA signals) and full-stack web.

TECHNICAL SKILLS

Trading	MQL5 Expert Advisors and indicators (.ex5 resources), MetaTrader 5 Strategy Tester, hedging accounts, DCA / grid recovery, hedge-on-reversal, equity-curve trading, ATR / regime-adaptive sizing, XAUUSD, Forex, indices, crypto
Signals & TA	SuperTrend, ADX, RSI, MACD, Bollinger / squeeze, Ichimoku, Stochastic, Donchian, smart-money concepts (BOS, order block, FVG), volume-spread analysis, multi-timeframe confirmation
Risk	Virtual balance, max-drawdown stop, profit lock, margin guard, spread-spike block, net-exposure cap, daily limits, consecutive-loss pause, persistent state across restarts
Python	Binance API, Streamlit, Telegram Bot API, n8n automation (built a real-time crypto monitor with MA-crossover signals and Telegram alerts)
Web & Product	Next.js 15, React 19, TypeScript, Prisma, NextAuth, Tailwind, Zustand, Zod, PayOS webhooks (HMAC-SHA256), Vercel, license-key generation

WORK EXPERIENCE

Independent Trading-Systems Developer

2026 – Present

- Develop, document and sell **B. Nexus Quantum** (v3.47) with a listing on MQL5 Market, a public website, community channels (Telegram, Zalo) and demo accounts tracked on MyFxBook. Runs the whole loop alone: strategy and risk engine, licensing, payments, guides, presets and support.

SELECTED PROJECTS

B. Nexus Quantum – MetaTrader 5 Expert Advisor

b-nexus-quantum.vercel.app

MQL5 · MetaTrader 5 · XAUUSD · Strategy Tester

- **Architecture:** 11 MQL5 modules (signals, entry and DCA, position management, smart trim, adaptive recovery, equity curve, dashboard, time and news filters, utilities, scoring), ~27K lines, 500+ configurable inputs, 3 proprietary indicators embedded as resources.
- **Signal engine:** 21 entry modes plus up to 3 entry and 3 DCA confirmation layers; **SmartTrend** scores 18 weighted layers (multi-timeframe, trend quality, momentum, zones, market structure, volume) against an adaptive threshold with severity veto, fakeout filter and breakout detection.
- **Position logic:** 5-tier DCA with 8 trigger modes and 4 lot modes, two independent BUY/SELL cycles with persisted state, hedge-on-reversal, zone recovery, 15 take-profit mechanisms (trailing, momentum burst, session- and regime-aware).
- **Smart Trim and Adaptive Recovery:** pairs BUY and SELL positions to cut drawdown at positive net profit (multi-match, partial closes, emergency mode); a 6-component sideways-market controller brakes DCA and caps net exposure.
- **Adaptivity:** 4-state volatility regime classifier with 3-bar confirmation, London/NY/Asia session scaling, ATR auto-scaling of TP, DCA and SL, automatic symbol-category detection (gold, forex, JPY pairs, indices, crypto, silver, oil) with digit-normalisation safeguards.
- **Operations:** on-chart dashboard and control panel, heartbeat monitor, account- and expiry-bound licence check, MQL5 Market DRM compatibility.

B. Nexus Quantum website – licensing, payments and docs

Next.js 15 · React 19 · Prisma · NextAuth · Zod · Vercel

- Next.js 15 / React 19 app (~19.5K lines TypeScript) with 12 Prisma models: users, orders, licences, linked trading accounts, trial requests, downloads, pricing plans, backtest images, payment webhooks and guide articles.
- PayOS payment webhook verified with HMAC-SHA256 that auto-issues licence keys; account link / unlink / change API, admin dashboards for orders, licences, users and pricing, public guides, backtest gallery and pricing pages.

EDUCATION & LANGUAGES

B.Eng. in Information Technology · Vietnam-Korea University of ICT (VKU), Da Nang — 2020 – 2025

- Languages: Vietnamese (native), English (professional working proficiency: technical reading and writing).
- Trading involves risk. Backtests and demo results are not a guarantee of future performance.